

Project: Integrating Machine Learning to Advance Your Research

Objective:

This project aims to leverage machine learning techniques covered in the course—such as regression, classification, support vector machines (SVM), neural networks (NN), clustering, and deep reinforcement learning (DRL)—to design, optimize, or enhance a system or process relevant to the student's research area.

While the primary focus may include advanced energy systems for planning and operation, students are also encouraged to explore other domains where ML can bring measurable improvements.

Course: SEEN5320 (Prof. Xiangyu Li)

Task:

- To design, optimize, or modify a system or process relevant to your research by applying machine learning techniques studied in the course. The scope of application can be either macro (e.g., transportation networks, manufacturing systems, environmental monitoring, healthcare operations) or micro (e.g., industrial equipment, sensor data analysis, small-scale robotics, individual decision-making processes).
- **Submit a technical report in PDF or Word, simulation/code/model, and a presentation in PPT.**

Learning Outcomes:

- Synthesize knowledge from the entire course to design a comprehensive machine learning-driven solution for a real or realistic problem.
- Present and defend the project findings, demonstrating technical and analytical skills in problem formulation, model selection, implementation, and evaluation.

Deadline:

Due before Week 12 (i.e., Apr. 29).

Presentation are scheduled on Weeks 12 & 13.

Grading Rubrics

Criteria for assessment	Performance level				
	Excellent	Good	Satisfactory	Pass	Fail
	10	8	6	4	2
Coverage of content (30% weighting)	Comprehensively addresses all key aspects of the topic; demonstrates a deep understanding of the context and task; includes a thorough literature review that critically assesses and integrates relevant sources, highlighting gaps and trends in existing research.	Covers most key aspects of the topic with minor omissions; shows a solid understanding of the context and task; includes a literature review that is adequate but may lack critical analysis or depth in exploring certain studies or trends.	Addresses the main components of the topic but lacks depth in several areas; demonstrates basic understanding of the context and task; literature review is present but limited in scope, relying on general sources without critical engagement.	Offers superficial coverage, focusing on obvious points while neglecting significant elements; shows a limited grasp of the context and task; literature review is minimal and lacks relevance, with major omissions of key studies or theories.	Fails to address essential content areas; demonstrates a misunderstanding of the task objectives and context, rendering the work largely irrelevant; literature review is absent or consists of non-scholarly sources, lacking any meaningful engagement with existing research.
Methodology (40% weighting)	Employs a well-structured methodology that is highly appropriate for the task. Demonstrates advanced understanding of relevant techniques, justifying choices with solid reasoning. All steps are documented thoroughly, allowing for reproducibility and transparency, and includes validation of results through appropriate testing or comparison with established benchmarks.	Utilizes a sound methodology that is mostly appropriate for the task; clearly articulates the main steps taken. Shows a good understanding of relevant techniques, with some justification for choices made. Documentation is clear but may lack some detail in specific areas. Validation of results is present but may be less comprehensive.	Describes a functional methodology that addresses the task but lacks depth in certain areas. Steps taken are somewhat clear, but there may be ambiguities or inconsistencies in execution. Justification for methodological choices is present but limited. Validation efforts are minimal or not robust, leading to potential questions about reliability.	Presents a poorly defined methodology that lacks clarity and appropriateness for the task. Significant gaps in the description of methods make it difficult to understand the process. Justification for methodological choices is weak or absent. Validation is either not attempted or very superficial, raising concerns about the credibility of the results.	Fails to provide a coherent methodology; lacks any meaningful description of methods used. There is no justification for choices made, and the process is not documented, rendering the work non-reproducible. Validation of results is absent, leading to a complete lack of confidence in the findings.
Critical analysis (20% weighting)	Demonstrates in-depth critical analysis that thoroughly evaluates evidence, arguments, and assumptions. Integrates insights from a variety of perspectives and sources, leading to a deep understanding of the topic.	Shows solid critical analysis that evaluates key evidence and arguments effectively. Integrates multiple perspectives but may not fully explore all implications.	Provides basic critical analysis that identifies strengths and weaknesses of the main arguments but lacks depth. Insights are mostly surface-level and do not draw from a wide range of sources.	Offers superficial analysis that focuses on obvious points without deeper examination. Relies on a narrow set of sources, resulting in limited insights.	Fails to provide any meaningful critical analysis; relies solely on descriptive summaries without evaluation. Insights are absent or irrelevant.
Creativity and novelty (10% weighting)	Exhibits a high degree of creativity and originality, presenting innovative ideas and approaches that significantly enhance the understanding of the topic.	Demonstrates creativity and originality in several aspects of the work, presenting interesting ideas and approaches.	Displays a moderate level of creativity, with some original ideas or approaches, but largely relies on existing concepts and frameworks.	Shows minimal creativity, primarily reiterating existing ideas without introducing new perspectives.	Fails to demonstrate any creativity or originality. The work is entirely derivative, lacking unique ideas or perspectives.

Final Grade Descriptors

Grades	Short Description	Elaboration on subject grading description
A (100-90)	Excellent Performance	Demonstrates a comprehensive grasp of subject matter, expertise in problem-solving, and significant creativity in thinking. Exhibits a high capacity for scholarship and collaboration, going beyond core requirements to achieve learning goals.
B (89-80)	Good Performance	Shows good knowledge and understanding of the main subject matter, competence in problem-solving, and the ability to analyze and evaluate issues. Displays high motivation to learn and the ability to work effectively with others.
C (79-70)	Satisfactory Performance	Possesses adequate knowledge of core subject matter, competence in dealing with familiar problems, and some capacity for analysis and critical thinking. Shows persistence and effort to achieve broadly defined learning goals.
D (69-60)	Marginal Pass	Has threshold knowledge of core subject matter, potential to achieve key professional skills, and the ability to make basic judgments. Benefits from the course and has the potential to develop in the discipline.
F (59-0)	Fail	Demonstrates insufficient understanding of the subject matter and lacks the necessary problem-solving skills. Shows limited ability to think critically or analytically and exhibits minimal effort towards achieving learning goals. Does not meet the threshold requirements for professional practice or development in the discipline.

Course AI Policy:

Generative artificial intelligence can be utilized to enhance coding and writing. Students should acknowledge its use in their work.

Communication and Feedback:

Assessment marks for individual assessed tasks will be communicated via Canvas within two weeks of submission. Feedback on assignments will include strengths and areas for improvement.

Resubmission Policy

N/A

Tools:

MATLAB, Python.

Academic Integrity

Students are expected to adhere to the university's academic integrity policy. Students are expected to uphold HKUST(GZ)'s Academic Honor Code and to maintain the highest standards of academic integrity. The University has zero tolerance of academic misconduct. Please refer to Regulations for Academic Integrity and Student Conduct for the University's definition of plagiarism and ways to avoid cheating and plagiarism.